THREAT HUNTING & APT DETECTION

PRESENTED BY GROUP 1

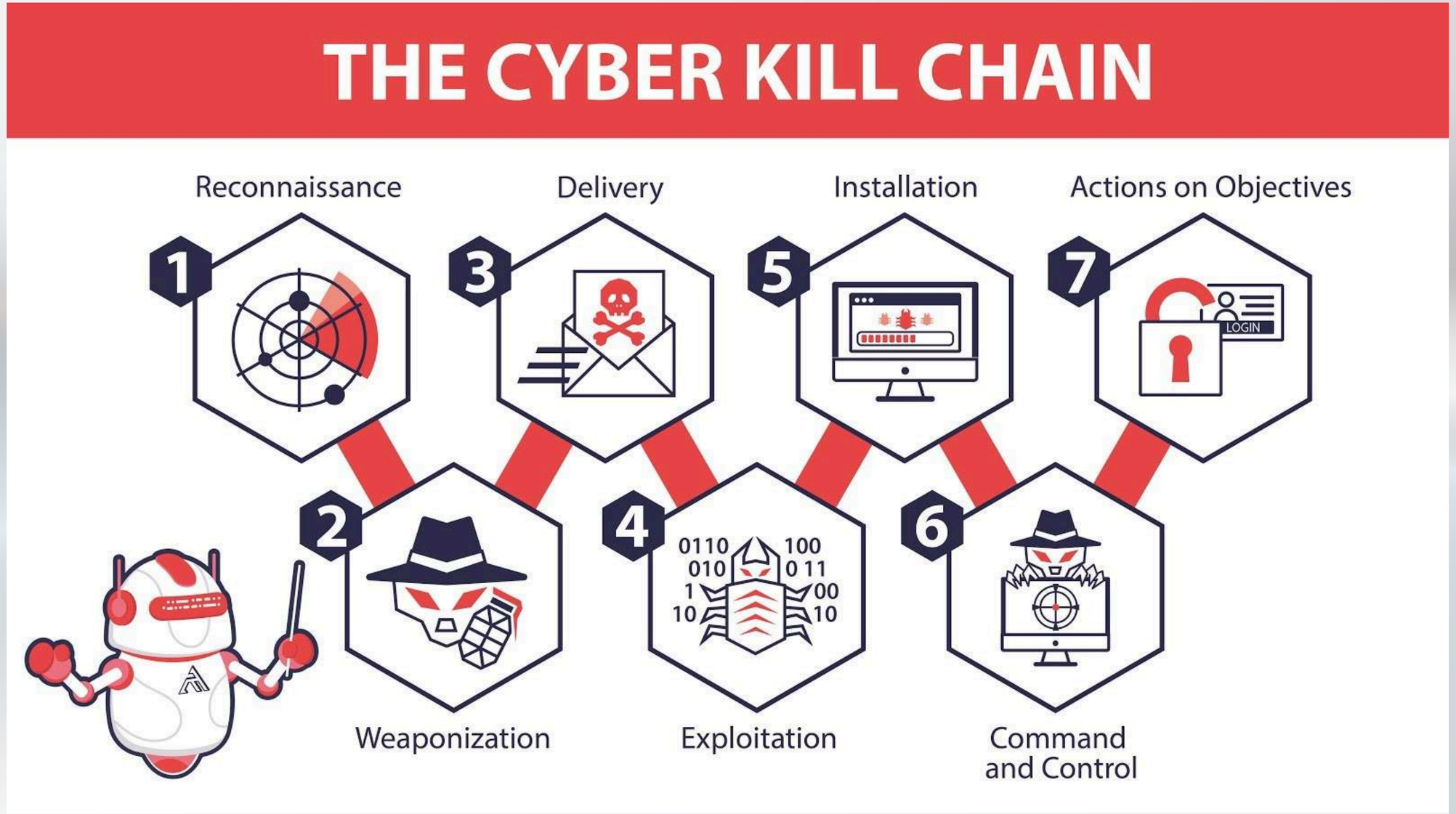
Fagr Hassan
Faten Mahmoud
Seif Amged
Youssef Hanna
Ismail Maher

What defines an APT?

- Persistent → long-term presence in network
- Targeted → specific organizations or sectors
- Stealthy → avoids detection
- Well-resourced → skilled teams & funding

Goal: espionage, data theft, or disruption

Cyber Kill Chain



Deception Technology

Definition:

- A security approach based on misleading attackers
- Uses fake assets to expose malicious behavior early

Deception Techniques

- Honeypots: Decoy systems that attract attackers
- Canary Tokens: Lightweight deception triggers

Reactive vs Proactive

- Reactive: investigate alerts after damage
- Proactive: lure attackers into traps

Honeypots as a Proactive Threat Hunting Tool

What is a Honeypot?

- A honeypot is a decoy system designed to attract attackers.
- It looks real, but is fully monitored and isolated.
- No legitimate user should ever access it

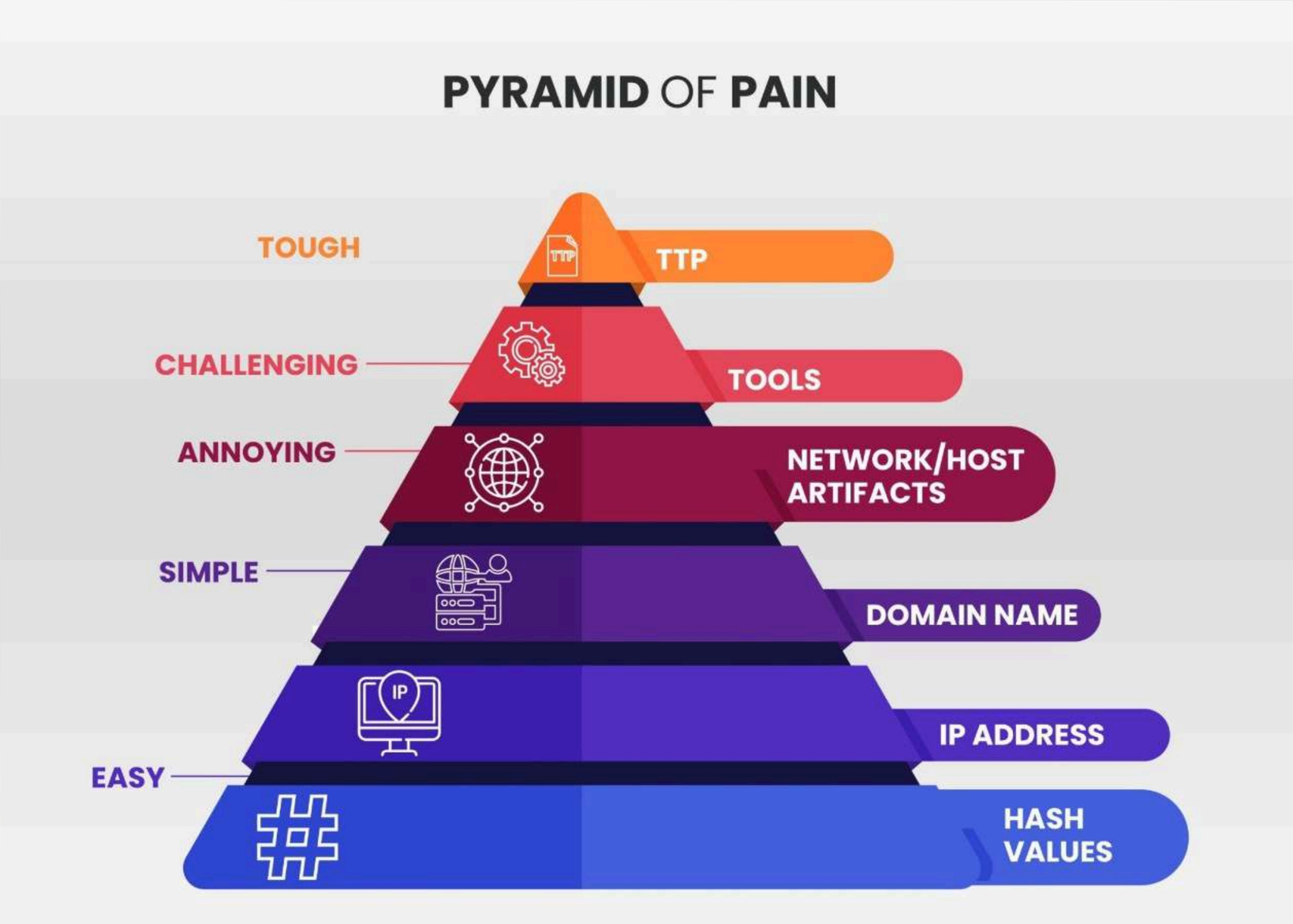
Why It Matters for Threat Hunting?

- Any interaction = 100% attacker activity
- Helps hunters detect reconnaissance and lateral movement
- Eliminates false positives

Types of Honeypots

- **Low-Interaction**
  - Simulated services
  - Easy to deploy
  - Low risk
- **High-Interaction**
  - Real operating system
  - Full attacker engagement
  - Used for deep analysis

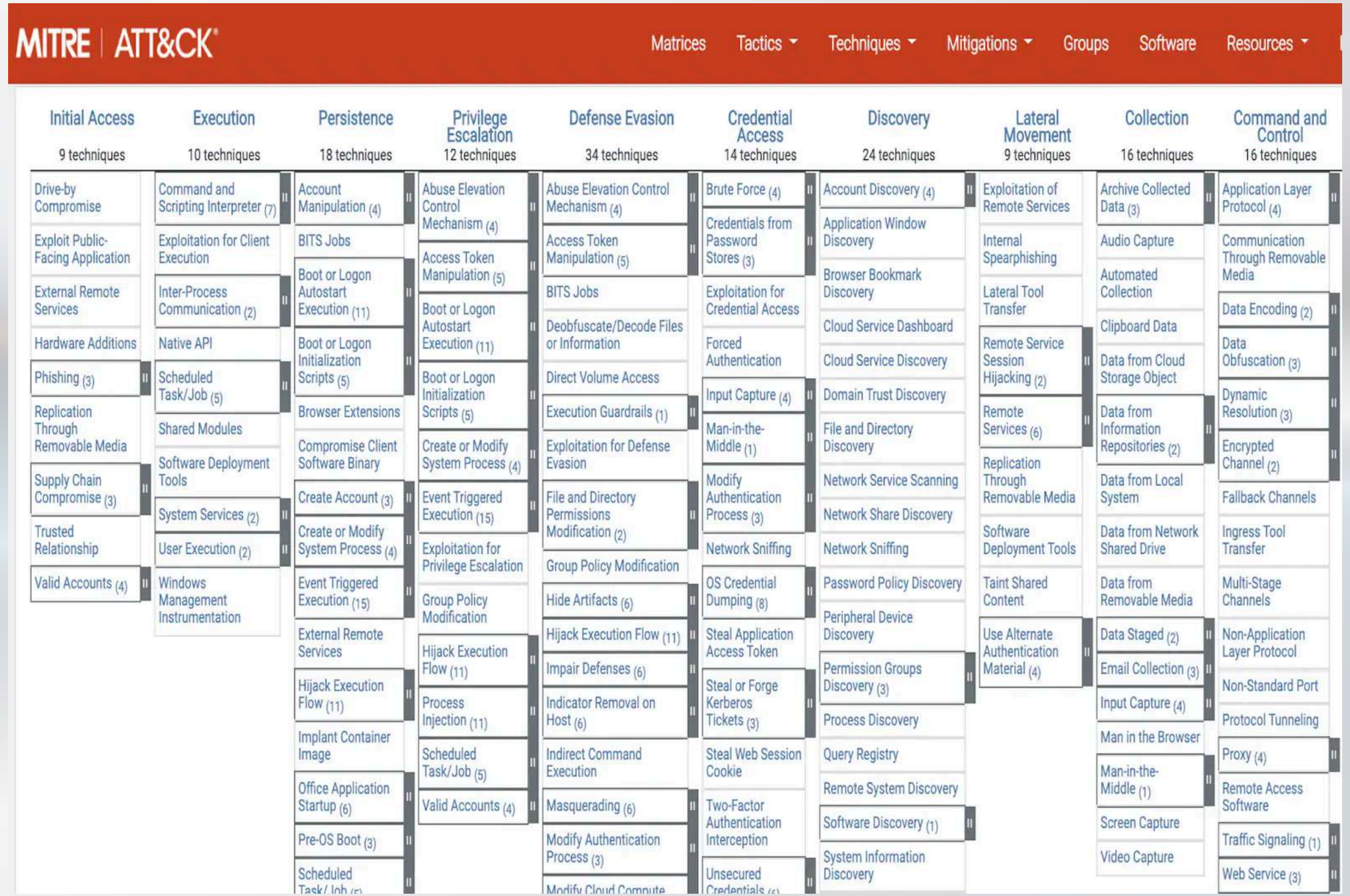
Pyramid of Pain



MITRE ATT&CK Framework

- Global knowledge base of attacker behavior
- Organized as a matrix
- Maps Tactics & Techniques (TTPs)
- Used by SOC & threat hunters

MITRE ATT&CK Framework



Connecting Honeypots to MITRE

- Honeypots act as behavioral sensors
- Every interaction reveals attacker TTPs
- Events are aligned with ATT&CK techniques
- Result:
  - Clear visibility of attacker tactics
  - Better threat profiling
  - Stronger response strategy

Detection & Analysis Tools

SIEM (Splunk / ELK)

- Centralized log collection
- Correlation across systems
- Detect attack patterns

Sysmon

- Process monitoring
- Registry tracking
- Endpoint behavior visibility

Memory Forensics (Volatility)

- Analyze RAM
- Reveal hidden malware
- Detect fileless attacks

Static vs Dynamic Analysis

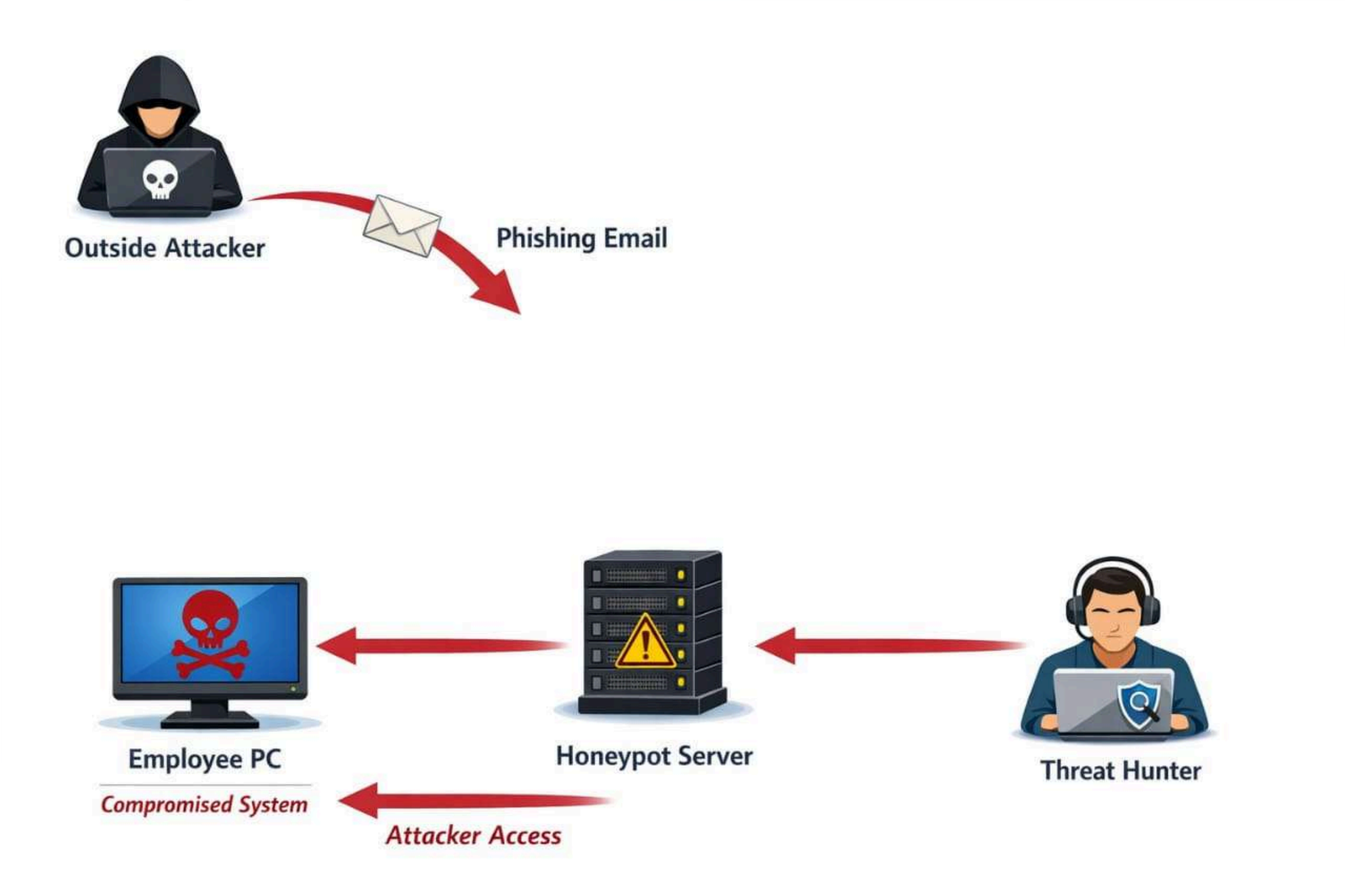
Static Analysis:

- Examine files without running
- Fast & safe
- Limited behavior insight

Dynamic Analysis:

- Run in sandbox
- Observe real behavior
- Detect hidden actions

From Phishing to Detection — Inside a Threat Hunter Investigation



DEMO

Conclusion

- APTs are stealthy and persistent
- Deception reveals attackers early
- Honeypots reduce false positives
- Threat hunting = proactive defense

“The best defense is a proactive offense.”

Questions?